Design and Analysis of Algorithms – Assignment 2 (CO2)

Complete Output of solution_code.py (Q1–Q7)

Q1 – Sorting

q1_sorting_results.csv

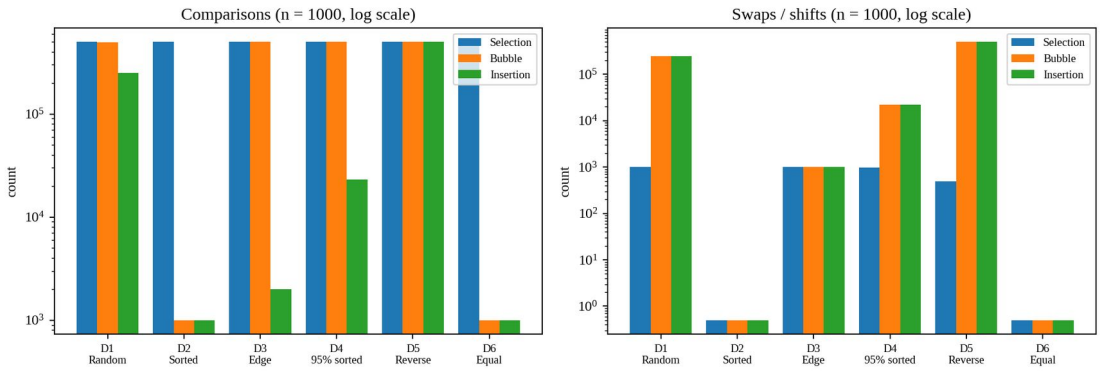
dataset	algorithm	n	comparisons	swaps_or_shifts	time_ms
D1 Random order	Selection	1000	499500	992	23.762
D1 Random order	Bubble	1000	496497	249484	47.516
D1 Random order	Insertion	1000	250479	249484	26.454
D2 Already sorted	Selection	1000	499500	0	22.711
D2 Already sorted	Bubble	1000	999	0	0.057
D2 Already sorted	Insertion	1000	999	0	0.088
D3 Edge: sorted + 1 top-priority order at end	Selection	1000	499500	999	23.33
D3 Edge: sorted + 1 top-priority order at end	Bubble	1000	499500	999	26.89
D3 Edge: sorted + 1 top-priority order at end	Insertion	1000	1997	999	0.18
D4 Scenario: 95% sorted + 5% new batch appended	Selection	1000	499500	978	23.98
D4 Scenario: 95% sorted + 5% new batch appended	Bubble	1000	498324	22134	28.828
D4 Scenario: 95% sorted + 5% new batch appended	Insertion	1000	23133	22134	2.227
D5 Reverse sorted	Selection	1000	499500	500	23.819
D5 Reverse sorted	Bubble	1000	499500	499500	57.316
D5 Reverse sorted	Insertion	1000	499500	499500	44.93
D6 All equal scores	Selection	1000	499500	0	23.131
D6 All equal scores	Bubble	1000	999	0	0.057
D6 All equal scores	Insertion	1000	999	0	0.159

q1_batch_sweep.csv

batch_size	percent_new	sel_comps	sel_swaps	bub_comps	bub_swaps	ins_comps	ins_shifts
1	0.1	499500	125	117999	125	1124	125
10	1.0	499500	839	486134	5574	6573	5574
50	5.0	499500	974	499310	29428	30427	29428
100	10.0	499500	989	498225	48847	49846	48847
250	25.0	499500	993	497289	108123	109122	108123
500	50.0	499500	994	498720	192288	193286	192288

q1_scaling_random.csv

n	sel_comps	sel_swaps	bub_comps	bub_swaps	ins_comps	ins_shifts
250	31125	243	30915	15123	15368	15123
500	124750	496	124735	60116	60608	60116
1000	499500	993	498972	247115	248108	247115
2000	1999000	1989	1994905	979491	981485	979491



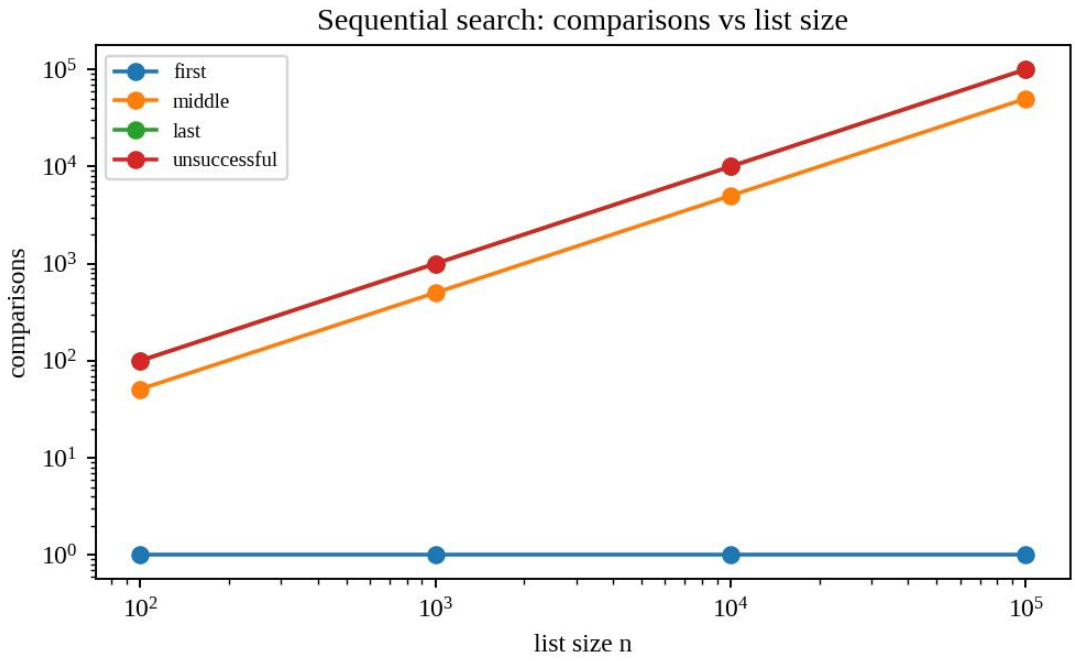
Q2 – Sequential search

q2_sequential_search.csv

n	scenario	index	comparisons
100	Successful - first position	0	1
100	Successful - middle position	50	51
100	Successful - last position	99	100
100	Unsuccessful (T-ID absent)	-1	100
100	Successful - average of 100 random keys		50.5
1000	Successful - first position	0	1
1000	Successful - middle position	500	501
1000	Successful - last position	999	1000
1000	Unsuccessful (T-ID absent)	-1	1000
1000	Successful - average of 1000 random keys		500.5
10000	Successful - first position	0	1
10000	Successful - middle position	5000	5001
10000	Successful - last position	9999	10000
10000	Unsuccessful (T-ID absent)	-1	10000
10000	Successful - average of 2000 random keys		5039.5
100000	Successful - first position	0	1
100000	Successful - middle position	50000	50001
100000	Successful - last position	99999	100000
100000	Unsuccessful (T-ID absent)	-1	100000
100000	Successful - average of 2000 random keys		49960.1

q2_boundary_model.csv

n_active_records	max_lookups_per_sec_at_10pct_core	worst_latency_ms
500	2725.8	0.04
2000	681.5	0.15
10000	136.3	0.73
50000	27.3	3.67
100000	13.6	7.34
500000	2.7	36.69



Q3 – String matching

q3_string_matching.csv

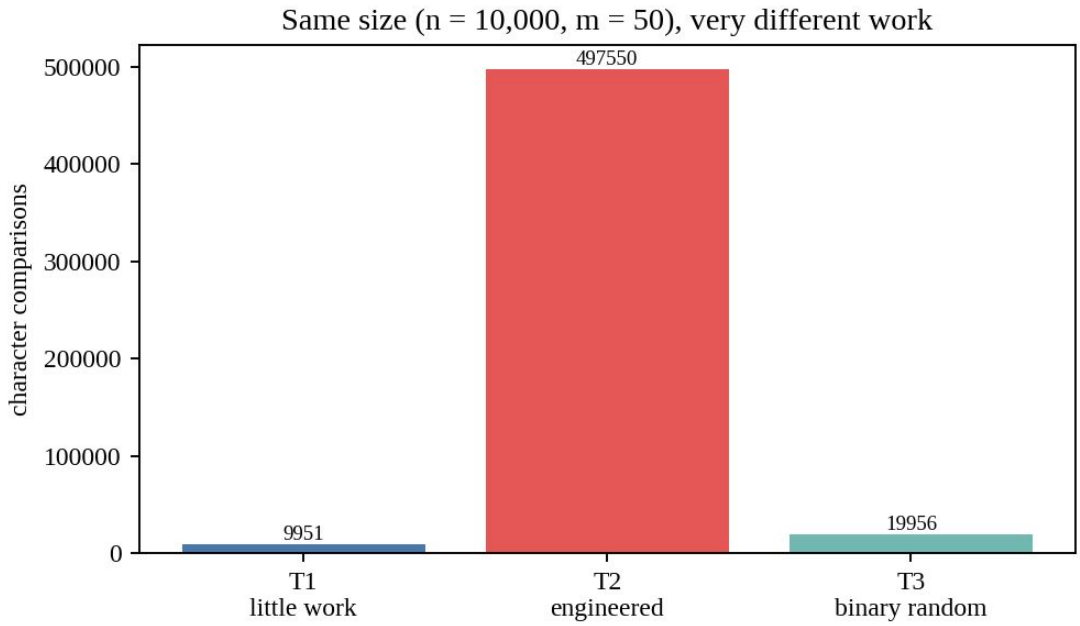
test	n	m	predicted_comparisons	observed_comparisons	matches
T1 Little work: first pattern char absent from text	10000	50	9951	9951	0
T2 Engineered: all-A text, pattern A^49 B	10000	50	497550	497550	0
T3 Own choice: random binary text, random binary pattern	10000	50	19902	19956	0

q3_alphabet_sweep.csv

alphabet_size	predicted	observed
2	19902	20052
4	13268	13254
26	10349	10384

q3_worst_case_pattern_length.csv

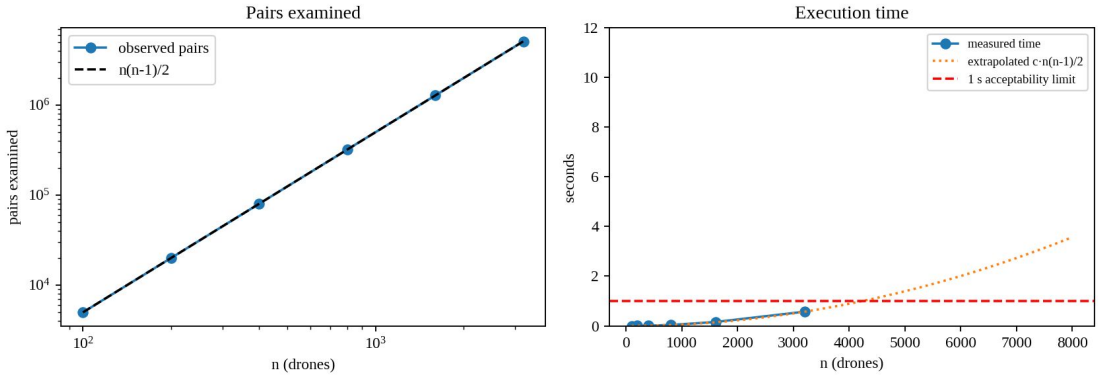
m	predicted_m(n-m+1)	observed
5	49980	49980
10	99910	99910
25	249400	249400
50	497550	497550
100	990100	990100
200	1960200	1960200



Q4 – Closest pair

q4_closest_pair.csv

n	pairs_examined	n(n-1)/2	distance_calculations	time_s
100	4950	4950	4950	0.00046
200	19900	19900	19900	0.00201
400	79800	79800	79800	0.00847
800	319600	319600	319600	0.03787
1600	1279200	1279200	1279200	0.15652
3200	5118400	5118400	5118400	0.57068



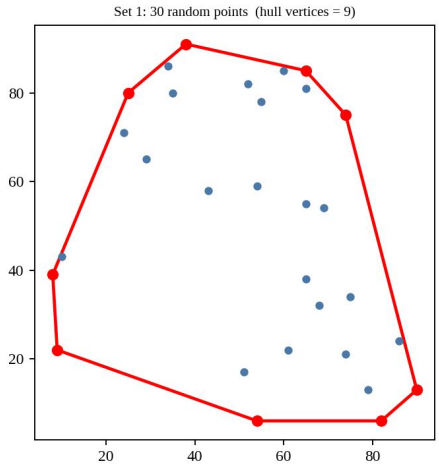
Q5 – Convex hull

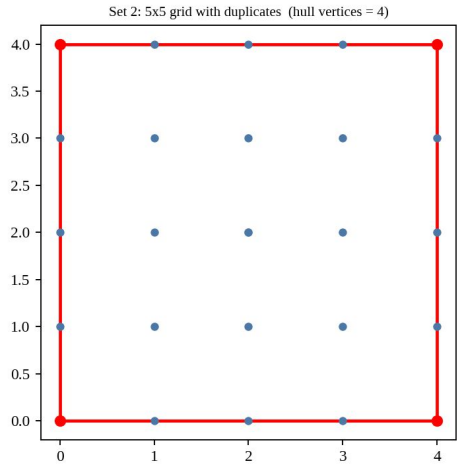
q5_hull_tests.csv

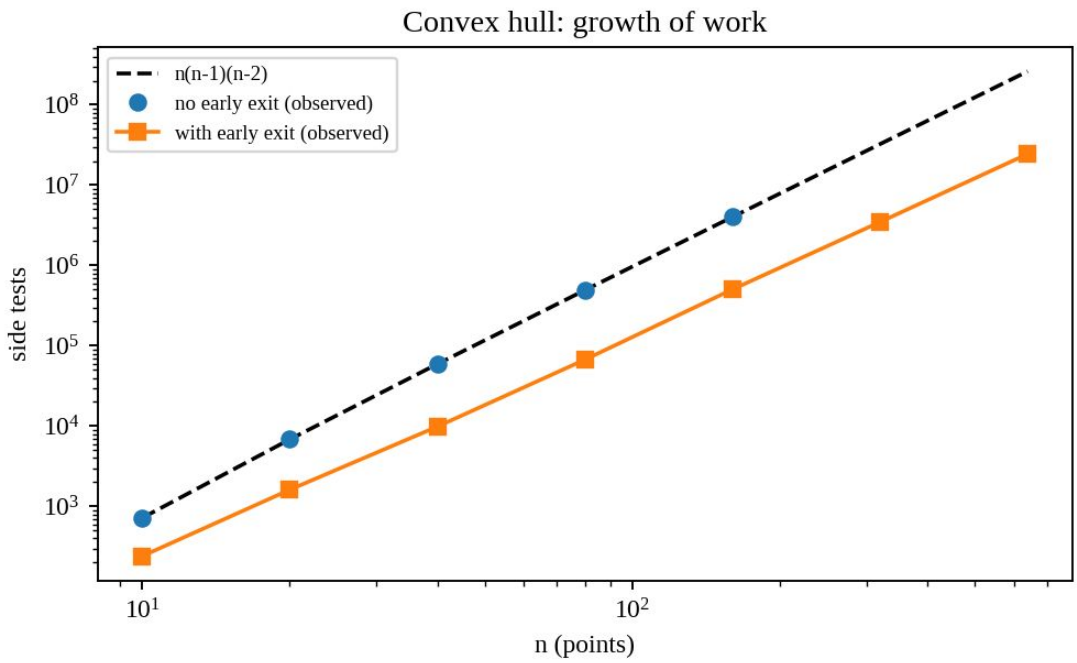
test	distinct_points	hull_vertices	side_tests	matches_reference
Set 1: 30 random points (boundary + interior)	30	9	4807	True
Set 2: 5x5 grid + duplicates (collinear boundary points)	25	4	2674	True
Triangle + 3 interior points	6	3	61	True
All points collinear	8	2	96	True
Parabola y=x^2 (all points on hull)	13	13	561	True
Two points only	2	2	0	True

q5_hull_growth.csv

n	hull_vertices	side_tests_with_early_exit	side_tests_without_early_exit	n(n-1)(n-2)	time_early_s	time_full_s
10	4	241	720	720	0.0001	0.0002
20	8	1616	6840	6840	0.0005	0.0015
40	9	9866	59280	59280	0.0024	0.0133
80	11	67254	492960	492960	0.0159	0.1008
160	20	504090	4019520	4019520	0.107	0.863
320	14	3478278		32461440	1.0444	
640	18	24454330		260916480	5.0062	







Q6 – Exhaustive search

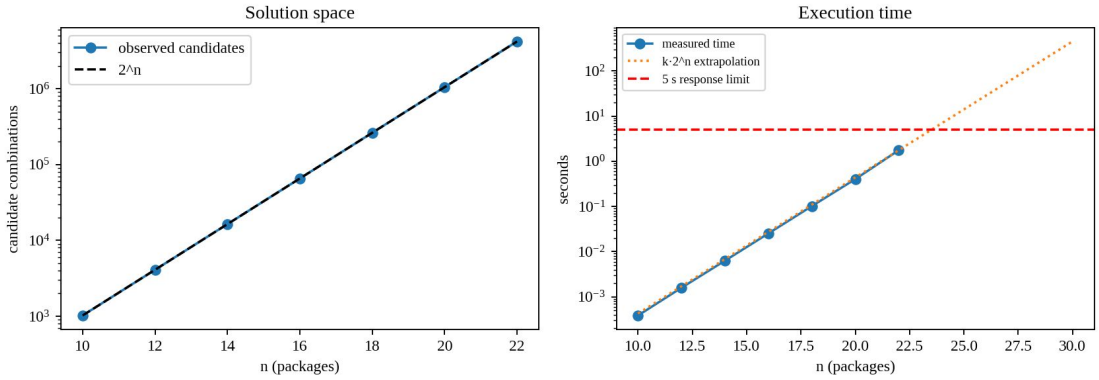
q6_small_instance_all_subsets.csv

subset	total_weight	total_value	feasible(<=26)
{}	0	0	True
{P1}	12	24	True
{P2}	7	13	True
{P1,P2}	19	37	True
{P3}	11	23	True
{P1,P3}	23	47	True
{P2,P3}	18	36	True
{P1,P2,P3}	30	60	False
{P4}	8	15	True
{P1,P4}	20	39	True
{P2,P4}	15	28	True
{P1,P2,P4}	27	52	False
{P3,P4}	19	38	True
{P1,P3,P4}	31	62	False
{P2,P3,P4}	26	51	True
{P1,P2,P3,P4}	38	75	False
{P5}	9	16	True
{P1,P5}	21	40	True
{P2,P5}	16	29	True
{P1,P2,P5}	28	53	False
{P3,P5}	20	39	True
{P1,P3,P5}	32	63	False
{P2,P3,P5}	27	52	False
{P1,P2,P3,P5}	39	76	False
{P4,P5}	17	31	True
{P1,P4,P5}	29	55	False
{P2,P4,P5}	24	44	True
{P1,P2,P4,P5}	36	68	False

{P3,P4,P5}	28	54	False
{P1,P3,P4,P5}	40	78	False
{P2,P3,P4,P5}	35	67	False
{P1,P2,P3,P4,P5}	47	91	False

q6_exhaustive_growth.csv

n	2^n_predicted	candidates_generated	feasible	tree_nodes	time_s	optimum_value
10	1024	1024	311	2047	0.0004	347
12	4096	4096	1079	8191	0.0016	388
14	16384	16384	4174	32767	0.0063	541
16	65536	65536	15721	131071	0.0253	506
18	262144	262144	59960	524287	0.1017	709
20	1048576	1048576	214253	2097151	0.4023	684
22	4194304	4194304	829309	8388607	1.7545	698



Q7 – Redesign

q7_original_vs_redesign.csv

n	original_nodes(2^(n+1)-1)	redesign_nodes_avg5	node_reduction_factor	redesign_complete_candidates	bound_pruned	infeasible_include_skipped	original_time_s	redesign_time_s
10	2047	22.6	90.6	1.6	6.4	7.6	0.0004	3e-05
12	8191	33.8	242.3	1.8	9.4	12.4	0.0015	3e-05
14	32767	34.2	958.1	1.4	9.6	13.2	0.0067	8e-05
16	131071	39.2	3343.6	1.6	11.0	15.0	0.026	3e-05
18	524287	61.2	8566.8	1.8	16.0	26.6	0.1354	4e-05
20	2097151	56.2	37315.9	1.0	16.0	23.2	0.5105	0.00011
22	8388607	74.8	112147.2	1.0	21.8	30.2	1.8529	9e-05

q7_redesign_larger_sizes.csv

n	redesign_nodes_avg5	redesign_nodes_max	redesign_time_s	original_nodes	original_time_estimated_s
25	57.6	70	0.0001	67108863	14.0
30	69.4	93	0.0001	2147483647	449.1
35	103.6	234	0.0001	68719476735	14372.6
40	135.8	195	0.0002	2199023255551	459924.2
50	124.0	149	0.0002	2251799813685247	470962428.3

q7_adversarial_instance.csv

n	redesign_nodes	original_nodes	fraction_of_original	time_s
12	2295	8191	0.28	0.0011
16	34517	131071	0.263	0.0166
20	576293	2097151	0.275	0.2709
22	1796087	8388607	0.214	0.9558

